TEXORA · LECTURE SHEET

Sewing Threads

A sewing thread must run at full machine speed, survive the needle, and then hold a seam for the life of the garment. This sheet covers the requirements, constructions, ticket numbers and selection of sewing threads.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 7 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. What a sewing thread is

A **sewing thread** is a smooth, even, twisted yarn — singles or plied — specially finished so that it forms stitches cleanly and holds two or more plies of material together in a seam. It differs from an ordinary weaving or knitting yarn in three ways: it is made and finished for *sewability* as well as strength, passing repeatedly through the needle eye and fabric at high speed without fraying, melting or breaking; it is sold by a commercial size system (the ticket number, Section 5) rather than by yarn count alone; and it is supplied on a package from which it unwinds smoothly without snarling.

2. Requirements of a good sewing thread

A thread strong in a tensile test can still sew badly. The full requirement set is:

- **Adequate seam strength and elasticity.** The seam — not just the thread in isolation — must carry the loads in wear, and the thread must extend slightly with the fabric under stress and recover, instead of snapping or cutting the fabric. A seam much stronger than the fabric is wasteful; a weaker seam fails first.
- **Evenness and smoothness.** Thick places jam in the needle eye and tension discs; thin places break. Low hairiness and a smooth surface reduce friction and lint build-up on the machine.
- **Controlled friction.** Friction must be low enough for smooth running and correct stitch formation, but not so low that stitches slip. Lubricated finishes (Section 6) set this balance.
- **Twist balance.** A lively, unbalanced thread snarls and kinks when slack, causing skipped stitches. Plied threads are twisted opposite to the singles twist so the finished thread is torque-balanced.
- **Heat resistance.** In high-speed sewing the needle becomes very hot from friction with the fabric; the thread in its eye must not melt, soften or lose strength — a serious limit for low-melting synthetics on heavy fabrics, and one reason lubricants also cool the needle.

- **Resistance in use.** Colour fastness to washing, light, perspiration and dry-cleaning, abrasion resistance in the seam, and, where needed, resistance to rot, chemicals, bleach or ultraviolet light. The thread must match the garment's care and service life, not just its colour on the day of making.
- **Appearance match.** Colour, lustre and thickness must suit the fabric — inconspicuous for a neat seam, or deliberately contrasting for topstitching.

> **A USEFUL RULE**
>
> Choose the **finest** thread that still gives the required seam strength: it needs a finer needle, makes a smaller needle hole, lies flatter in the seam and usually sews more smoothly.

3. Constructions of sewing thread

3.1 Spun thread

Staple fibres — cotton, polyester staple, or blends — are spun into singles yarns, two or three of which are plied together and finished. Spun polyester is the workhorse apparel thread: good strength, abrasion resistance and reliable sewing. Spun cotton is soft and heat-tolerant but weaker and less abrasion-resistant; surviving a hot needle well keeps it in use for garment dyeing and work on natural-fibre fabrics.

3.2 Core-spun (corespun) thread

A continuous filament core — usually polyester — is wrapped during spinning with a sheath of staple fibre, usually polyester or cotton. The core supplies most of the tenacity; the sheath supplies a soft, fibrous surface, good loop formation and needle-heat protection for the core. Polyester-core / polyester-sheath (poly-poly) is the premium general-purpose apparel thread; polyester-core / cotton-sheath is used where a cotton face is wanted, for example for garment-dyed goods. Figure 1 shows the structure.

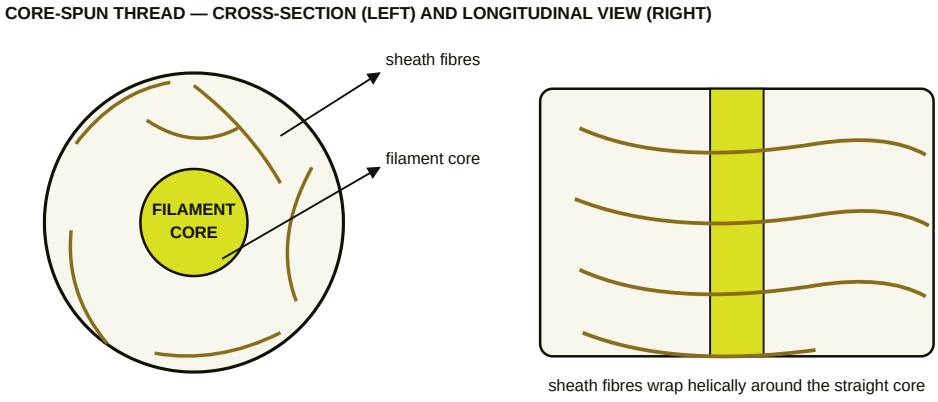


Figure 1 — Core-spun sewing thread. A straight continuous-filament core carries most of the load; staple fibres wrapped around it form the sewing surface.

3.3 Continuous-filament thread

Several filaments are combined, usually plied and bonded. Filament polyester or nylon is the strongest construction for a given size, very smooth and abrasion-resistant — used for footwear, upholstery, luggage and heavy seams. Its smoothness is also a risk: stitches can slip if tension is not controlled.

3.4 Monofilament thread

A single thick filament, usually nylon, used as one strand. Nearly transparent, so one thread serves many colours — used for hemming, quilting and budget upholstery. It is stiff and harsh against skin, so it is avoided next to skin and for quality seams.

3.5 Textured thread

Textured (bulked) polyester or nylon filament: crimped, loopy filaments with high bulk and stretch. Soft, with excellent cover, it is the standard looper thread for overlock and coverstitch seams on knitwear. Weaker per size than flat filament, it is not used as needle thread in stressed seams.

CONSTRUCTION	STRENGTH FOR SIZE	SURFACE / FEEL	SEWING BEHAVIOUR	TYPICAL USES
Spun staple	Moderate	Soft, matt, fibrous	Reliable; some lint	General apparel seams
Core-spun	High	Soft sheath, strong core	Excellent; low breakage	Quality apparel, denim
Continuous filament	Highest	Smooth, sometimes lustrous	Very smooth; watch stitch slip	Shoes, bags, upholstery
Monofilament	Moderate	Stiff, transparent	Harsh; heat-sensitive	Blind hems, quilting
Textured	Lower	Bulky, soft, high cover	Excellent cover in loopers	Overlock / coverstitch loopers

4. Fibre choices

- **Polyester** — the dominant thread fibre: high tenacity, excellent abrasion and chemical resistance, good colour fastness, low shrinkage, strength kept wet. Available as staple, filament, core-spun and textured.
- **Cotton** — soft, matt, heat-tolerant at the needle, and it swells when wet, helping to seal needle holes in waterproof cotton fabrics. Weaker than polyester; used where its handle, dye behaviour or heat

tolerance matters, and as a core-spun sheath.

- **Nylon (polyamide)** — very high strength and abrasion resistance with high elasticity; standard for bonded heavy-duty threads. More heat- and UV-sensitive than polyester, which has taken many of its outdoor uses.
- **Rayon / viscose** — bright, with excellent colour range; used mainly for embroidery threads, not load-bearing seams (low strength, especially wet). **Special fibres** — aramids for heat-resistant seams, PTFE-type and water-soluble threads — follow the same logic at much higher cost.

The fibre rule is matching: thread should match the fabric in strength, shrinkage, care behaviour and service environment. Polyester thread in a garment-dyed cotton garment will not take the dye and shows as undyed stitching — a fibre-choice fault, not a dyeing fault.

5. Ticket numbering

Threads are bought by **ticket number**, a commercial size number printed on the package, linked to the metric count (Nm) of the finished (resultant) thread by the approximate rule:

$$\text{Ticket number} \approx 3 \times \text{Nm of the resultant thread, where Nm} = 1000 / \text{tex}$$

Two cautions keep this rule honest. First, the ticket is a *commercial* number, not a precise measurement: makers round it to standard values, and constructions at the same ticket can differ slightly in real tex — check the maker's stated tex when seam strength or costing must be exact. Second, its direction is opposite to tex: a **higher ticket means a finer thread** (it follows Nm), while a higher tex means a coarser thread. Common apparel tickets run from about 20 (heavy topstitching, denim) through 40–60 (general seaming) to 100–150 and finer (light fabrics).

> **WORKED EXAMPLE — TICKET NUMBER FROM TEX**
>
> A resultant sewing thread has a linear density of 60 tex.
> Nm = 1000 / 60 = 16.7.
> Ticket ≈ 3 × Nm = 3 × 16.7 = 50.0, so the thread is sold as approximately **ticket 50**.

6. Thread finishes

- **Lubrication.** A controlled application of silicone or wax-based lubricant lowers friction, reduces needle heating and breakage, and evens out stitch tension. Too little gives friction, heat and breaks; too much can stain light fabrics and make stitches slip — so the level is specified, not left to chance.
- **Bonding.** A resin coating glues a filament thread's strands together so it sews as one strand without untwisting or fraying. Bonded nylon and polyester are the standard heavy-duty threads for shoes, bags and upholstery.

- **Glazing.** A starch/resin and wax finish, polished under friction, gives cotton thread a hard, smooth, lustrous surface — the classic glazed cotton for leather and heavy goods; it is stiff, so it is not used where a soft seam is needed. Other finishes include mercerising (lustre, strength, dye uptake), water-repellent and soft finishes.

7. Selecting a thread — fabric, seam and needle

Selection proceeds in order: (1) seam stress sets the required seam strength; (2) fabric weight sets thread size — light fabrics need fine threads and needles, heavy fabrics coarser ones; (3) fibre is matched to fabric care and dyeing route; (4) construction is chosen — core-spun for stressed apparel seams, spun for general work, bonded filament for heavy goods, textured for loopers; (5) needle is matched to thread: the thread must pass freely through the eye and lie in the groove — too thick, it frays and breaks; too fine in a coarse needle, stitches wander and the needle leaves large holes. Finally, test-sew the actual fabric before bulk production: most machine-blamed faults are thread–needle–fabric mismatches.

Glossary

Sewability — ability to run on a machine without breaks, skipped stitches or seam faults. **Resultant tex** — linear density of the finished plied thread. **Ticket number** — commercial thread size; higher ticket = finer thread. **Core-spun thread** — filament core wrapped with a staple sheath. **Bonded thread** — filament thread resin-coated so its strands sew as one. **Glazed thread** — thread with a hard polished finish. **Textured thread** — bulked filament thread with high cover, used mainly in loopers. **Looper thread** — thread supplied to an overlock or coverstitch machine's loopers.

Key points

• A sewing thread needs seam strength and elasticity, evenness, controlled friction, twist balance and needle-heat resistance — not tensile strength alone.
• Five constructions — spun, core-spun, filament, monofilament, textured — chosen by strength, surface and seam type.
• Polyester dominates; cotton, nylon and rayon each hold specific niches.
• Ticket ≈ 3 × Nm of the resultant thread (Nm = 1000 / tex); higher ticket = finer thread; confirm real tex from the maker.
• Finishes set sewing behaviour: lubrication controls friction and needle heat; bonding holds filament strands together; glazing gives a hard polished surface.
• Select by seam stress → fabric → fibre → construction → needle match, then test-sew the combination.

Practice

1. A resultant thread is 45 tex. Calculate its Nm and approximate ticket number, and state whether it is finer or coarser than ticket 50.
2. Choose a construction and fibre for: (a) a stressed denim side seam, (b) overlock loopers on a T-shirt, (c) a leather bag seam. Justify each in one sentence.
3. Explain why a thread can pass a tensile test yet break repeatedly on a high-speed machine. Name three causes.
4. A buyer asks for a "thicker thread, ticket 150 instead of ticket 40". Correct the request: explain what ticket 150 actually is relative to ticket 40, and what to ask for if a coarser thread is wanted.

Answer key

1. Nm = 1000 / 45 = 22.2. Ticket ≈ 3 × 22.2 = 66.7, about **ticket 67**. Ticket 67 is **finer** than ticket 50 — higher ticket means finer.
2. (a) Core-spun polyester — filament-core tenacity with a soft, sewable sheath. (b) Textured polyester — bulky, soft, high-cover in the looper, where coverage matters more than strength. (c) Bonded filament nylon or polyester — highest strength and abrasion resistance, the bond preventing ply separation.
3. Tensile strength is measured slowly on a straight length. On the machine the thread also fails by abrasion in the needle eye and guides, needle heat softening or melting it, thick places jamming and snapping, and unbalanced twist snarling into a knot that breaks.
4. Ticket numbers run opposite to thickness: ticket 150 is much **finer** than ticket 40. For a coarser thread, ask for a **lower** ticket number (or higher resultant tex) and confirm the maker's tex.
